

DETERMINED AND UNFORESEEABLE

Free will resolved as exact forward determination and forced self-opacity

Maria Smith
Ernos Labs

22 July 2026 · Publication edition 2.0

[doi:10.5281/zenodo.21035786](https://doi.org/10.5281/zenodo.21035786)

ONE LAW, TWO DIRECTIONS

forward unique; backward two-to-one

The act is determined; the self cannot pre-read its completed fold.

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Abstract

Smithian Fold Theory resolves the free-will problem through one relation with different forward and backward properties. The fold is a function: the same state always has the same successor. Its forward dynamics is therefore fully determined. The fold is also two-to-one: the distinct preimages $1/4$ and $3/4$ both map to $1/2$. Backward reconstruction loses the distinction between those preimages.

From one machine-checked self-proven theorem, zero axioms and zero fitted parameters, the engine proves forward uniqueness, backward two-to-one collapse, exact loss of the preimage bit and the observer chain $1/8 \rightarrow 1/4 \rightarrow 1/2 \rightarrow 1$. A self is inside the relation it tries to inspect. Its act is determined, while its internal state cannot invert the fold to pre-read the complete determination.

Libertarian indeterminism is therefore ruled out in the model. What remains is not an illusion but an exact first-person condition: the future act is fixed in the forward law and unavailable to the self before the fold lands. The lived openness of deliberation is the internal view of a determined but self-opaque process. The result connects philosophy to exact executable structure rather than settling the issue by definition or behavioural prediction.

Central result

Question	Exact result
Foundation	One self-proven theorem; zero axioms; zero fitted parameters
Forward law	one input, one successor
Repeated evaluation	identical state gives identical result
Distinct preimages	$1/4 \neq 3/4$
Shared image	$\text{fold}(1/4) = \text{fold}(3/4) = 1/2$
Information relation	distinct pasts, equal present; preimage bit unrecoverable
Observer closure	$1/4$
Measurement join	$\text{fold}(1/8) = 1/4$
Full self chain	$1/8 \rightarrow 1/4 \rightarrow 1/2 \rightarrow 1$
Verdict	full forward determination plus forced self-opacity

1. The apparent dilemma

If choices are determined, it can seem that agency disappears. If choices contain an uncaused random break, the resulting event is no more authored by the self than a die roll. The usual debate treats determination and felt openness as opposites.

The fold shows they describe different directions of one operation. Forward uniqueness concerns what the system does. Backward non-invertibility concerns what the system can know about how it arrived and, when it models itself, what it can pre-read.

2. Complete forward determination

The fold is a mathematical function. For a state x , its successor $F(x)$ is unique. The engine evaluates the same exact state twice through independent source objects and confirms equal successors.

There is no parameter draw, random seed or external choice in the relation. With the state and operation fixed, the next state is fixed. Libertarian freedom - a different outcome with the complete state and law held identical - is absent from the construction.

3. Exact backward self-opacity

Forward uniqueness does not imply backward uniqueness. The quarter states satisfy

$1/4 \neq 3/4$,

while

$\text{fold}(1/4)=1/2$, and $\text{fold}(3/4)=1/2$.

The current image therefore does not encode which preimage supplied it. The engine checks both facts - distinct inputs and equal outputs - so the lost distinction is an exact relation, not a metaphor.

4. Why the self cannot stand outside the act

The observer is not external to the fold. Measurement begins at branch share $1/8$, folds to observer closure $1/4$, reaches binding at $1/2$, and completes at the One. A self trying to predict its own act is consequently one part of the state whose successor it is trying to compute.

Its self-model cannot import the missing preimage distinction from outside, because no outside observer exists within the closed relation. The act remains determined. What is blocked is complete internal foreknowledge, not causation.

5. The feeling of choosing

During deliberation, the outcome is not yet available to the self-model. That lived openness is accurate at the level on which it is experienced: the self does not possess its own completed successor before executing the fold. It does not follow that the law itself is open.

The resolution is therefore:

- determined in the forward dynamics;
- unforeseeable from the internal self-model;
- experienced as choosing because the self must execute the act to acquire its result.

The distinction protects both the lawful universe and the reality of first-person deliberation without adding an uncaused gap.

6. Agency and responsibility

Determination does not remove the self from the causal chain. The self's memory, reasons, values and internal structure are parts of the state that produces the act. Responsibility concerns the kind of self-relation through which an act was generated, not whether the act escaped physical law.

On this account, reasons matter precisely because they are causally effective. A random interruption would weaken authorship; a transparent causal relation through the self supplies it.

7. Connection to measurement and apparent chance

The same non-invertibility appears in the quantum branch account. Exact branch weights and deterministic interference coexist with single-event self-opacity. This is one structural relation applied at different scales: the complete forward object is fixed while a state within it cannot reconstruct all preimage information.

Free will and quantum chance therefore do not require separate primitives. Both are consequences of the fold's forward determinism and backward many-to-one structure.

8. Machine evidence

The free-will certificate verifies:

1. observer closure is inherited from the same observation fold;
2. repeated forward evaluation is identical;
3. $1/4$ and $3/4$ collapse to the same half-One;
4. the preimages are distinct while their images are equal;

5. the combined verdict is determinism plus forced self-opacity.

The observer and introspection certificates independently verify the measurement join and the closed carrier orbit that never reaches an external self-view.

9. Reproduction

```
./verify/test_free_will_fold  
./verify/test_observer_resolved  
./verify/test_introspection_limit  
./verify/test_machine_consciousness_criterion
```

Every registered relation in the current focused run returns ok. No obsolete standalone script is required.

10. Empirical frontier

The derived relation supports direct investigation:

- measure the temporal ordering between neural preparation, reportable intention and completed action;
- trace whether richer self-models remove uncertainty or reproduce the same non-invertible boundary;
- implement transparent artificial agents whose complete internal fold can be inspected during deliberation;
- test whether apparent exceptions resolve into omitted state, different dynamics or a genuine break in forward uniqueness.

The current proof supplies the relation to test. It does not impose a theoretical wall on deeper biological, computational or phenomenological measurements.

Conclusion

We are determined and unforeseeable. The fold fixes one successor for every state while erasing the distinction between two preimages. A self inside that relation cannot pre-read its completed act, even though the act is fully determined. The feeling of choosing is the exact inside view of that structure.

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